

News-Driven Liquidity Dynamics in WTI Crude Oil Futures

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THE RESOLUTION GAP

News moves markets. The open question is when.

- Media sentiment predicts price pressure and trading volume (Tetlock, 2007), but almost all of that work is at **daily** resolution.
- The reaction to a release is concentrated within **minutes to hours** (Ederington & Lee, 1993). A daily bar averages that away.
- WTI is a good place to look: OPEC policy, geopolitics, the weekly EIA release and macro announcements all arrive at a **known** time, in a deeply liquid market.
- Kilian & Vega (2011) find **no** response of daily energy prices to daily macro news.

A null at daily frequency does not rule out an effect at hourly frequency.

WHAT WE ASK

RQ1. Lag structure

At what lag, if any, does news sentiment exert its strongest effect on WTI trading liquidity?

RQ2. Directional asymmetry

Does the liquidity response differ between bearish and bullish news, and if so, in what sense?

Liquidity is measured by **log trading volume** (activity), the **Amihud ratio** (price impact), and the **Parkinson range** (volatility). Two phases answer these: an interpretable baseline, then a richer model that corroborates it.

PHASE 1

An interpretable baseline

THE DATA COLLECTED

News from GDELT, market data from Yahoo Finance, on a common hourly grid.

Stage	Articles
Raw GDELT records, eight oil-market queries	51,948
After deduplication and English filter	16,326
Body retrieved (~80% success) and regex-filtered	7,756
Title-only fallback where the body failed	5,934
Phase 1 modelling set	13,690

Window: March 2024 to February 2026, against **11,219** hourly WTI records.

HOW FINBERT PROCESSED THESE ARTICLES

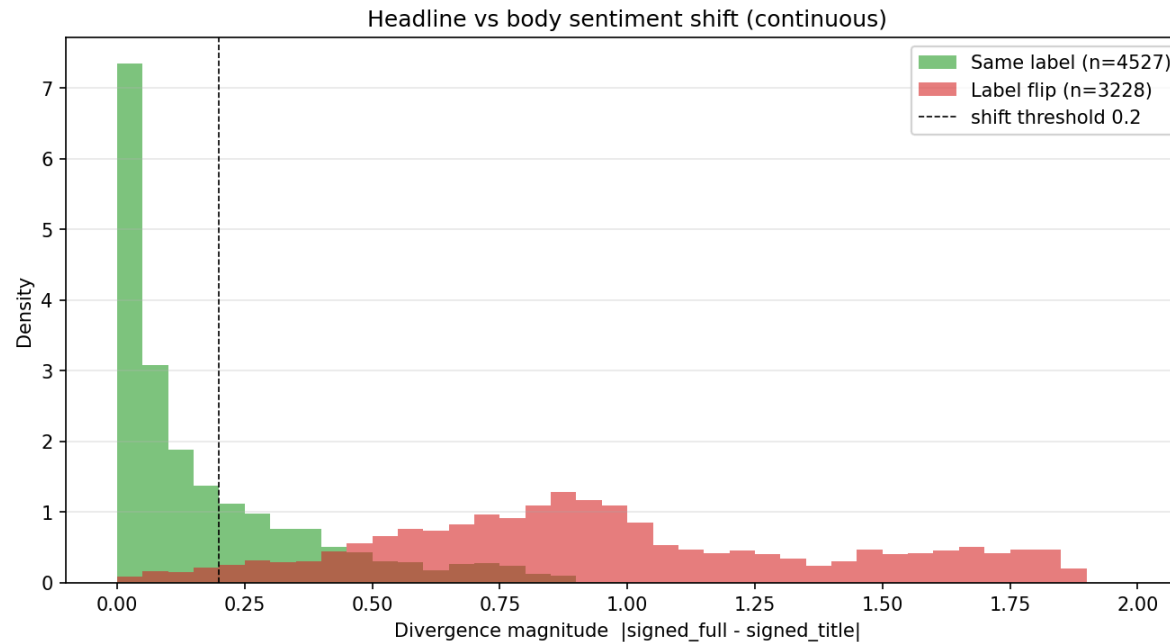
FinBERT is a BERT variant fine-tuned on 339,750 individual corporate and financial documents. It is applied to the articles only, never to market data.

- Each article is scored **twice**: on the **title alone**, and on the **title plus body** (truncated to 512 tokens, title kept at the front).
- The model returns three class probabilities per article: **P(positive), P(neutral), P(negative)**.
- **We keep the probabilities, not the argmax label**: a 0.95-bearish article should not count the same as a 0.51-bearish one.

One real article, "**Oil up 4pc for the week after solid demand forecasts**":

Input	P(positive)	P(neutral)	P(negative)	Label
Title only	0.93	0.03	0.03	positive
Title + body	0.02	0.01	0.97	negative

HEADLINE BIAS: TITLES ARE NOT ARTICLES



Scoring the same article twice isolates the input: **41.6%** of articles flip label, and the mean signed shift is **-0.09**. Headlines lean **more bullish** than the articles beneath them. For phase 1, title and title + body articles are considered.

PINNING NEWS TO TRADING HOURS

News publishes at any hour. Trading does not.

- **Ceiling rule.** An article at 14:23 is assigned to the **15:00** bar, never 14:00. The 14:00 bar was already forming before the news existed, so using it would let news explain a candle that predates it.
- **Forward-assignment.** Off-hours news (overnight, weekends, OPEC announcements) are carried to the next available trading hour rather than dropped. It is the most market-moving subset of the corpus.
- The gap between publication and assigned hour is kept as a control, separating **contemporaneous news (< 2h)** from **forward-assigned news ($\geq 2h$)**.

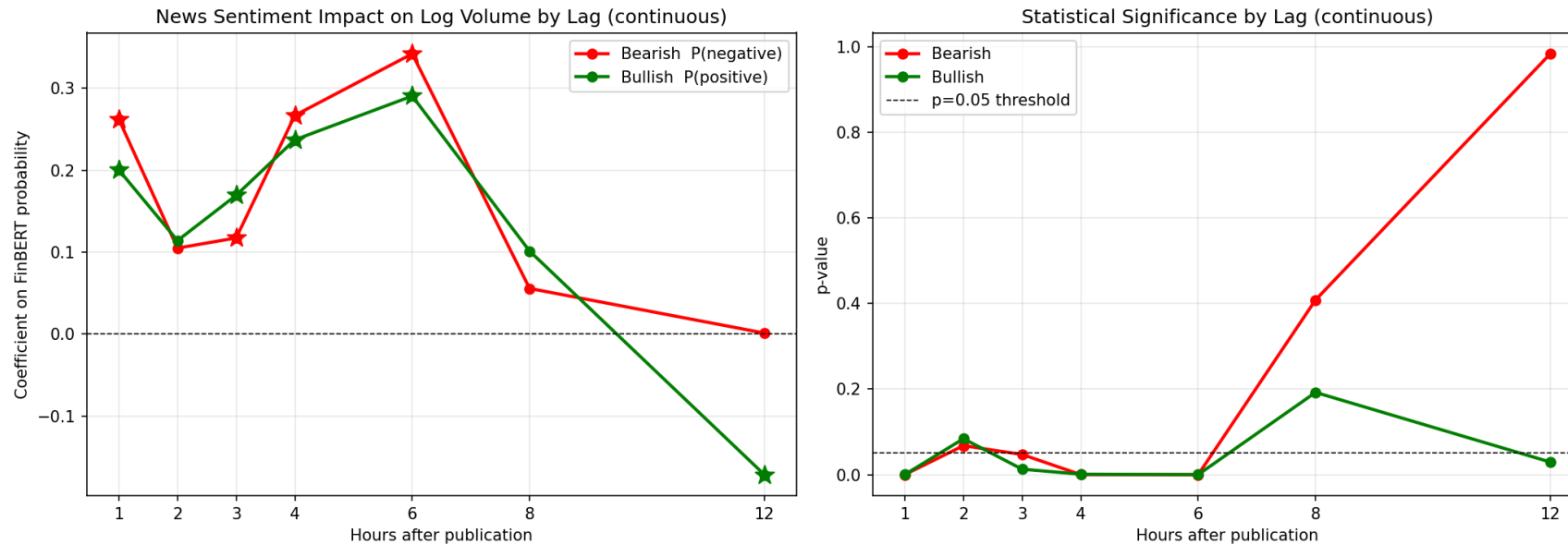
THE LAG REGRESSION

$$\log \text{volume}_{t+k} = \beta_0 + \beta_1 P(\text{neg})_t + \beta_2 P(\text{pos})_t + \varepsilon_t$$

- **Why ordinary least squares.** Nothing to tune and nothing to overfit: each β reads directly as the volume response to a maximally confident article. Neutral is the omitted reference, since the three probabilities sum to one.
- **Two coefficients, not one signed score.** A single signed regressor would force $\beta_1 = -\beta_2$, assuming the response is symmetric. Whether it is symmetric is exactly RQ2.
- **One regression per lag, $k \in \{0, 1, 2, 3, 4, 6, 8, 12\}$,** traces the effect over time. If news propagates with delay, the coefficients should **rise to a peak, then decay**, which is exactly what the next slide plots.
- **What it cannot do.** R^2 will be small: news is one driver of hourly volume among many. The finding is the **pattern** across lags, not the explained variance.

PHASE 1 · RESULTS

THE RESPONSE PEAKS AT +6 HOURS



RQ1: the effect builds, peaks sharply at **+6h** ($\approx 41\%$ more volume than a neutral hour), and decays to insignificance by +8h. **RQ2:** bearish exceeds bullish at every lag that carries the effect, by $\approx 18\%$ at the peak.

PHASE 1

WHAT THIS BASELINE COULD NOT DO

- **One sentiment axis.** Three classes, with no magnitude, event type, entities, or certainty.
- **A lexical filter.** The regex accepted long off-topic text and rejected short substantive briefs.
- **A sparse signal.** Over half of all hours carry no news; an exploratory VAR by consequence could not identify anything with significant precision and was abandoned.
- **No macro controls, and pre-war data only** (the corpus stopped in February 2026).

Each of these drives a specific Phase 2 change.

PART TWO

Phase 2: richer features, expressive model

WHAT WE SET OUT TO FIX

Each limitation of the baseline sets one task.

- **Get more data.** Extend the corpus past February 2026, and add macro controls so news is not carrying variance it never owned.
- **Extract richer features.** Replace three sentiment classes with a structured schema from an LLM.
- **Calibrate that extraction.** With no human ground truth, check it across model families before trusting it.
- **Change the model.** Sparse, event-driven news with many features and several horizons needs more than a single-lag regression.

The next slides take these in order.

WHAT WE CHANGED ABOUT THE INPUTS

Three changes before any modelling.

- **A bigger corpus.** Five new queries (Iran sanctions, Saudi production, China demand, Russia exports, supply disruption) and a re-scrape through May 2026: **22,795 articles**, now spanning the 28 February 2026 war onset.
- **Macro covariates.** DXY and VIX added as hourly controls. The Phase 1 regression explained under **0.3%** of volume variance, so letting the model absorb broad market movement leaves a cleaner residual for news.
- **Richer news features.** Using an LLM to extract features of news, using a schema v1, cross-calibrating against other LLM, then creating the schema v2.

WHAT THE SCHEMA REVISION CHANGED

Both extractions of "**Russian Sanctions Drive Oil Prices Higher**", before and after the revision:

Three channels added. `price_direction` dropped as redundant, `event_type` now salience-ordered, `usable` now required.

Schema v1

```
{
  "sentiment_score": 0.75,
  "magnitude": 0.85,
  "price_direction": "bullish",
  "event_type": "geopolitical",
  "certainty": 0.9,
  "time_horizon": "short_term",
  "entities": ["Russia", "US",
              "OPEC+", ...]
}
```

Schema v2

```
{
  "usable": true,
  "sentiment_score": 0.85,
  "supply_impact": -0.9,
  "demand_impact": 0.0,
  "risk_premium": 0.8,
  "magnitude": 0.85,
  "certainty": 0.9,
  "event_type": ["geopolitical",
                 "supply"],
  "time_horizon": "short_term",
  "entities": ["United States",
               "Russia", ...]
}
```

THE COMPOSITE FAILED, SO WE DECOMPOSED IT

No human ground truth was available, so we validated by cross-model agreement: the same 30 articles scored by Haiku and by a GPT-family model.

Metric	v1 schema	v2 schema
sentiment_score correlation	0.39	0.88
Sign disagreements	4 / 13 (31%)	1 / 14 (7%)
supply_impact / demand_impact / risk_premium	—	0.94 / 0.96 / 0.82

At $r = 0.39$ the disagreements concentrated on **geopolitical** events: one number was carrying two judgments, since a supply threat is bad news yet bullish for price. Splitting it into three channels (Kilian, 2009) fixed it, and **the composite improved without being modified**.

WHAT WE FED THE TRANSFORMER

Article-level features are aggregated to the hourly grid just as Phase 1, then passed to a Temporal Fusion Transformer.

- Continuous features are hour-averaged; entity flags take the maximum; categoricals take the highest-magnitude article of that hour; news-free hours get a `no_news` category.
- **71 canonical entity flags**, normalised so that "Tehran" and "Iranian" both map to Iran, entering as a binary vector per hour.
- `event_type` and `time_horizon` as **learned embeddings** rather than integer codes.
- **TFT v2**: 48-hour encoder, three targets, four horizons (+1, +3, +6, +12h), 60/20/20 split with the war onset **inside the test set**.

PREDICTION AGAINST A PERSISTENCE BASELINE

Target	+1h	+3h	+6h	+12h
log_volume	+46%	+60%	+67%	+71%
amihud	+43%	+43%	+45%	+45%
price_range	−45%	−24%	−14%	−3%

- Volume and Amihud beat persistence at every horizon, and the margin **grows with the horizon**.
- `price_range` fails, but only in the war regime; on the pre-war slice it matches persistence.

Trained on moderate volatility, the model reverts to the mean in a regime it never saw. A clean regime-extrapolation failure.

WHAT THE MODEL USES

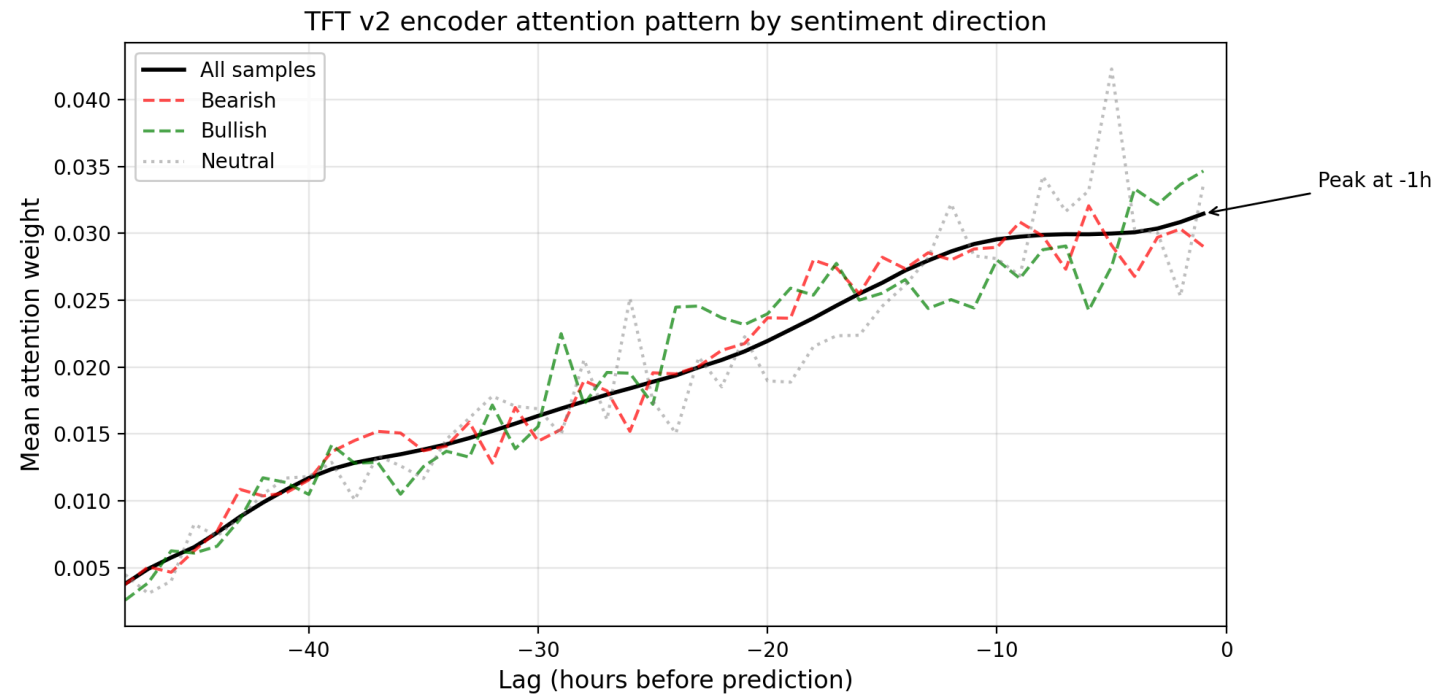
Rank	Feature	Weight
1	vix	0.188
2	supply_impact	0.121
3	ent_oman	0.113
4	demand_impact	0.055
6–10	ent_japan · ent_eu · ent_iran · ent_china · ent_algeria	0.017 – 0.014

Six of the top ten are entity flags, and they are economically legible: Oman and Iran around Hormuz, Japan, the EU and China as importers. The two channels together carry **17.6%**, comparable to VIX alone, while the composite `sentiment_score` does not appear at all.

Exact ranks shift on retraining. What is stable is the feature type: risk and salience over price direction.

PHASE 2 · RESULTS

WHERE THE MODEL LOOKS



Attention rises toward recency and peaks at **-1h**. Split by direction, **bearish-sentiment hours peak at -6h** while bullish peak at **-1h**: the same horizon the Phase 1 regression found, reached by an entirely different method.

PART THREE

Discussion

DISCUSSION

ANSWERING THE TWO QUESTIONS

RQ1. Settled, and settled twice

A single-lag linear regression and a 48-hour attention-based forecaster share no assumptions, and both place the response in the **+6 to +12 hour** window. The response is gradual, not instantaneous.

RQ2. Real, but it must be stated precisely

Phase 1 finds a robust bearish-over-bullish asymmetry in **marginal sensitivity**. Phase 2's sixteen tests of **average predicted volume** find nothing. These are different quantities, not a failed replication: a larger marginal sensitivity can coexist with equal group means. The model instead keys on **risk and salience**, not direction.

LIMITATIONS AND WHAT COMES NEXT

Limitations

- Validation rests on model agreement rather than expert human annotation ($n = 30$).
- `price_range` does not survive the regime boundary.
- One commodity, two years, one structural break, and a public web corpus rather than the institutional feeds traders read.

Next steps

- A small expert-annotated anchor set (~200 articles) to calibrate the calibration.
- Carry tone **and** price sentiment as parallel features, since their disagreement may itself be a signal.
- Multi-regime training, and cross-commodity replication.

MAIN FINDING

Liquidity responds to risk, not sentiment direction

The market does not trade more because the news is bad. It trades more because the news raises uncertainty about supply.

In oil, that uncertainty is often, against intuition, bullish for price.